

ARPITA BHATTACHARYA

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I am a Masters in Mathematics from Lund University, Sweden, highly motivated and ready to apply my experience in Numerical Analysis, Mathematical modelling, etc. to computation and applied fields in science and technology. My analytical approach in research has enabled me to solve challenging problems during the course of my education. My extracurricular activities and interests, especially music, have created the understanding of the value of social connect and harmony. I am looking forward for an opportunity to enhance knowledge that can help me attain growth and success through learning.

SKILLS

Software Programming	Python, R, C++, Java	★★★★☆
Simulation	MATLAB	★★★★☆
ML/AI	PyTorch	★☆☆☆☆
Data Processing/Visualization	Tableau	★★★★☆
Documentation	Markdown, Latex, MS Office	★★★★☆

PROJECT AND RESEARCH EXPERIENCE

MASTER THESIS (Supervisor: Prof. Erik Wahlén)

February 2025 – March 2026

Title: *Singularities in Steady Water Waves using Analytic Continuation*

Study of the analytic structure of steady two-dimensional water waves using tools from complex analysis, conformal mapping, and analytic continuation. The focus was on the study of the singularities in analytically continued wave profiles and how these influence the regularity and convergence properties of free-surface water wave representations.

PROJECTS (MASTER AND BACHELOR COURSES)

September 2023 – January 2025

Numerical Methods for Partial Differential Equations (Under guidance of: *Prof. Robert Klöf*, Lund University, Sweden)

Projects based on Elliptic partial differential equations, investigation of their properties using DUNE and DUNE-FEM in Python. The results were presented at the end of the course.

Numerical Methods for Differential Equations (Under guidance of: *Prof. Tony Stillfjord*, Lund University, Sweden)

Implementation of projects in MATLAB. The results were presented at the end of the course. The projects used concepts such as: Euler method, Adaptive Runge-Kutta methods, Lotka-Volterra equation, Van der Pol problem, 2-point BVP solver, Beam equation, Sturm-Liouville eigenvalue problem, Schrödinger equation, Diffusion equation, Advection equation, Convection diffusion equation, Viscous Burgers' equation.

Mathematical Modelling (Under guidance of: *Prof. Joachim Hein*, Lund University, Sweden)

Implementation of projects in Python and Microsoft Excel. The results were presented at the end of the course. The projects helped learn modelling process in its various stages: problem formulation, analysis, calculations, simulations and feedback.

Project 1: Optimization project: Textile distribution

Project 2: Analysis of Predator-prey modelling

Numerical Linear Algebra

Implementation of project tasks/assignments in Python and MATLAB, using concepts such as: Gram-Schmidt orthogonalization, QR-factorization, Householder transformations, Givens rotations, Image Compression using SVD, Hilbert Matrices, LUdecomposition, hyperlink matrices, GMRES.

ElGamal Cryptography

Implementation of the ElGamal cryptosystem in Python; submitted report demonstrating the working of the algorithm, including snippets from the Python code. The implementation was tested through challenging exercises and presentation of the solutions during the course.

Elliptic Curve Cryptography (Under guidance of: *Dr. Weiyi Zhang*, University of Warwick, United Kingdom)

A research essay on the cryptography and applications using Abstract Algebra and Number Theory.

SUBJECT MATTER EXPERT (FREELANCE)

February 2023 – April 2024

Chegg (Advanced Mathematics, *Managed Network Expert*, Bengaluru, India)

Knowledge sharing and Expert level problem resolutions for university and high school mathematics students.

EDUCATION

Lund University Lund, Sweden	MASTER IN MATHEMATICS (Numerical Analysis)	Sep 2023 – Mar 2026	Grade: G
University of Warwick Coventry, United Kingdom	BACHELOR IN MATHEMATICS	Sep 2019 – Aug 2022	Grade: Hons.
National Public School Bengaluru, Karnataka, India	HIGH SCHOOL	Jun 2005 – Mar 2019	Final Score: 91%
Centre of Fundamental Research and Creative Education (CFRCE) Bengaluru, Karnataka, India	RESEARCH EDUCATION	Apr 2017 – Aug 2019	GPA: 3.7

EXTRA-CURRICULAR

International Student Ambassador

September 2023 – March 2026
Lund University, Lund, Sweden

- Responsible for tasks designed to help prospective students in their decision-making process.
- Be a representative of the University.
- Answer questions and provide information via the Unibuddy platform and Live events.
- Create content for blogs and social channels of the University.

Company Host at ARKAD (Career Fair)

October 2023
Lund University, Lund, Sweden

- Be the Point of contact between ARKAD and visiting companies in the career fair. Help order resources and materials, set up booths, personally assist the company representatives.

Warwick Student Volunteer

January 2020 – March 2020
University of Warwick, Coventry, United Kingdom

- Helped primary school children learn mathematics through fun games, building their confidence in math concepts and making it fun as part of project 'Fun with numbers'.

National Centre for Human Settlements and Environment (NCHSE)

April 2018
Madhya Pradesh, India

- As part of "Integrated Watershed Management Programme" of the NCHSE, survey health, hygiene and sanitation awareness in the villages within the districts of Madhya Pradesh, India

OTHER INTERESTS/HOBBIES

Music

INSTRUMENT

- Sitar – Hindustani Classical

DANCE

- Hindustani Classical – Kathak, Bharat Natyam
- Other styles – Contemporary, Street, Hip-hop

Reading

FICTION, SCIENCE, TECHNOLOGY, PHILOSOPHY